

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Welding

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R00	25/07/22	Issued for Interdisciplinary Check
R01	25/07/22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-007

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviations	Meaning
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
AWS	American Welding Society
CS	Carbon Steel
DEP	SPDC- Design and Engineering Practices
DP	Design Pressure
DWT	Drop Weight tear
FEED	Front End Engineering
HAZ	Heat affected zone
FCAW	Flux Cored Arc Welding
HV 10	Vickers Hardness (10 kg load)
GMAW	Gas Metal Arc Welding
GTAW	Gas Tungsten Arc Welding
ISO	International Organisation for Standardization
IF	Incomplete Fusion
IFD	Incomplete Fusion Due to Cold Lap
IIW	International Institute of Welding
IP	Inadequate Penetration Without High-Low
IPD	Inadequate Penetration Due to High-Low
IQI	Image Quality Indicator
MMscf/d	Million standard cubic feet per day
NDT	Non-Destructive Testing
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SPDC	Shell Petroleum Development COMPANY Nig. Limited
SMAW	Shielded Metal Arc Welding
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

Site and Environmental Conditions

Refer to Basis of Design Document EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

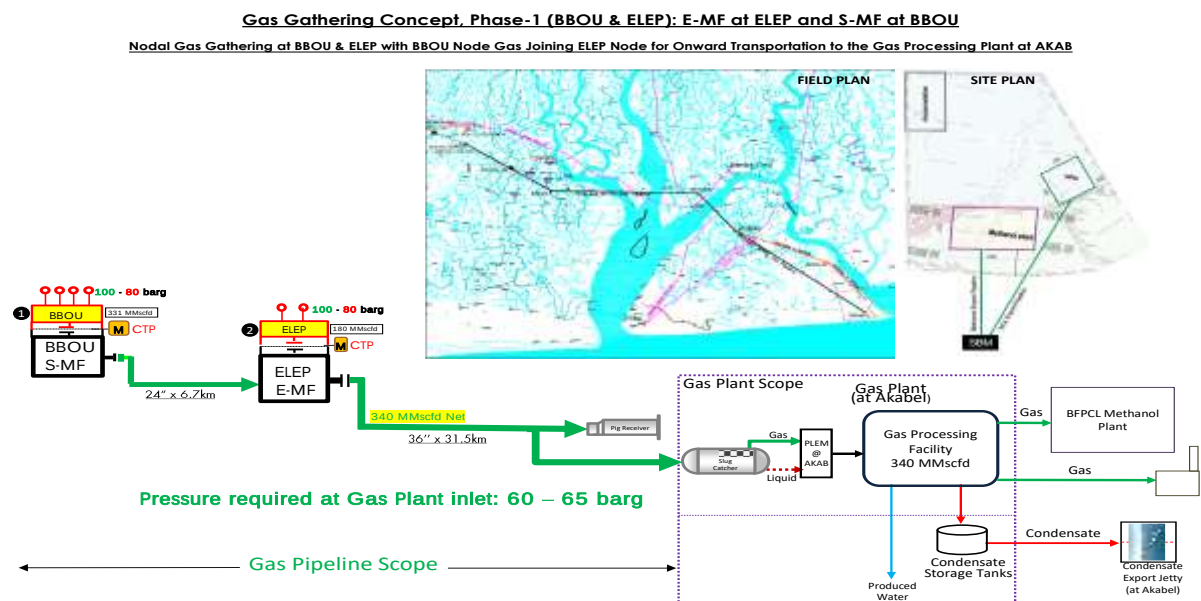


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This Specification covers the minimum requirements for welding, heat treatment, and non-destructive examination of pressure containing piping and/or equipment such as: pressure vessels, heat exchangers, etc installed in process facilities governed by ASME B31.3. These requirements also apply to welds attaching skirts, brackets, lugs and other non-pressure parts to such equipment. It is intended to apply to works on fabrication sites such as construction sites or fabrication yards, as well as in CONTRACTOR's workshops or for Vendor packages (skid assembly, etc.).

This document should be read in conjunction with DEP. 61.40.20.30-Gen and ANSI/API Std 1104

2. Regulations, Codes and Standards

The latest versions of the regulations, codes, and standards and extant amendment or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- COMPANY's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the COMPANY for resolution.

2.1. Project Technical Documents

Key Project Technical Specifications are listed below.

REF	DOCUMENT NUMBER	TITLE
1	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
2	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
3	EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Class Specification

2.2. Shell Specifications

The following DEPs are applicable:

REF	DOCUMENT NUMBER	TITLE
1	DEP. 61.40.20.30-Gen	Welding of Pipelines & Related Facilities (Amendments / supplements to API Std 1104)
2	DEP. 31.40.20.37-Gen	Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019)
3	DEP. 82.00.10.10-Gen	Project Quality Assurance
4	DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline (February 2017)

2.3. International Codes and Standards

REF	DOCUMENT NUMBER	TITLE
1	ASME B31.8	Gas Transmission and Distribution Piping Systems
2	ANSI/API Std 1104	Welding of pipelines and related facilities
3	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
4	API 5L	Specification for Line Pipe
5	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
6	API RP 5LW	Recommended Practice for Transportation of Line pipe on Barges and Marine Vessels
7	API RP 5L1	Recommended Practice for Railroad Transportation of Line Pipe
8	ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.
9	ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing
10	ASTM E 1815	Standard Test Method for Classification of Film Systems for Industrial Radiography
11	ISO 5579	Non-destructive testing - Radiographic examination of metallic materials by X-rays and gamma rays - Basic rules
12	ISO 15614	Specification and qualification of welding procedures for metallic materials
13	ASME Section VIII	Boiler and Pressure Vessel Code Divisions 1 and 2
14	ASME section II - Part C	Boiler and Pressure Vessel Code Section II - Specifications for Welding Rods, Electrodes, and Filler Metals

3. Line Pipes and Piping Material Data

The characteristics of the material involved are those listed and specified in the project piping material classes document EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005.

4. Welding Qualifications

4.1. General

- Welds shall have a minimum of two (2) passes whatever are the welding process and the weld geometry.
- Welding procedures without filler metal are prohibited.

4.2. Welding Procedures Specifications (WPS)

- The CONTRACTOR shall submit in his tender evidence of his ability to qualify welding procedures for each piping or equipment. This evidence shall be based on previous experience and qualification test results.
- All welds including weld repairs shall be supported by a relevant WPS. All WPS shall be in writing and shall be qualified and certified in accordance with the applicable code. All supporting PQR shall be witnessed, or at least certified, by a recognised inspection authority.
- All WPS, including their qualification test records (PQR) shall be submitted to the COMPANY for approval prior to starting fabrication. All test results shall be from specimens representative of the final condition of the equipment, including heat treatment(s) where applicable.
- Each WPS shall be identified by a unique number. For pressure vessels and other equipment, each WPS shall also be identified on a welding map.
- The WPS shall state the actual consumable or flux trade name and type as well as the code designation.
- Procedure tests for all welding processes shall be qualified using the same welding consumables and flux as will be used on the actual item. A change from one Manufacturer's consumable or flux to that from another Manufacturer, or change of type or grade from the same Manufacturer, shall require requalification for the following materials:
 - a. Carbon steel materials in sour wet service
 - b. Carbon steel materials subject to post weld heat treatment with impact test requirements
 - c. Any materials in cryogenic service (i.e. at temperature below minus 46°C)
 - d. Duplex stainless steels
 - e. Nickel alloys.
- For inert Gas Tungsten Arc (GTAW) and Inert Gas Metal Arc Welding (GMAW or FCAW), the qualification records shall include the composition and flow rate of inert gas shielding and backing when used. For Submerged Arc Welding (SAW), the qualification record shall include the current arc intensity and voltage, and the travel speed for each weld run.

- For Gas Metal Arc process (GMAW or FCAW), the records shall include the power sources, inductance setting, electrode diameter and extension, current and voltage, wire feed rate and travel speed.
- Tack welds shall be made to a qualified welding procedure that has been approved by the Engineering Company and by approved welders. Tack welds shall be completely removed prior to welding. Fit-up lugs and strong backs and other temporary attachments shall be attached using a qualified procedure approved by the Engineering Company and shall be welded by approved welders.
- Procedure qualifications for weld overlay deposits shall include the complete chemical analysis of the overlay. Specimens for chemical analysis shall be taken on the final weld layer. The weld metal chemistry shall be within the nominal range specified for the alloy.
- Qualification of overlays on tubes and fittings shall be made on same materials of the smallest size intended to be qualified and in the same welding position.
- The procedure qualification tests shall include:
 - f. Dye penetrant examination of the completed weld
 - g. Side bend tests and longitudinal face bend tests for weld metal soundness examination. Excessive cracking shall be cause for rejection. Cracks shall not exceed four per specimen nor shall they exceed 1.6 mm in each length. Cracks corners are not considered.
 - h. Photomacrograph.

4.3. Welders and Welding Operators

- Welders and welding operators shall be qualified in accordance with the applicable Code prior to fabrication. Records shall be submitted to the COMPANY for approval. Alternative qualifications may be accepted subject to the COMPANY approval.
- Welding operators and welder qualifications shall be valid for one year from last qualification performance date, beyond which requalification is required.
- However, a validity period of two years may be considered if sufficient evidence is shown that the related welder has regularly used the same welding process and materials with no interruption exceeding six months.
- Welders and welding operators are subject to requalification when the quality of their work, during fabrication, appears to be below the requirements of this Specification.

4.4. Welding production tests

- When required, production tests representative of each material and welding procedure for main seams shall be performed according to contractual documents requirements.
- Test coupons shall be of sufficient length to allow for all mechanical tests to be taken.
- Test coupons shall be welded attached to the part being welded where possible and

shall reflect the actual welding and heat treatment conditions of the piping or equipment.

- Mechanical tests and acceptance criteria shall be the same as required for welding procedure qualifications.

5. Welding Consumables

5.1. Filler Materials and Fluxes

5.1.1. Selection of filler materials and fluxes

- Welding consumables or combination of them used for the work shall produce welds with strength and ductility properties (especially notch toughness) in the final heat-treated condition (if applicable) at least equal to the minimum specified equivalent properties of the base material. For welding carbon steel materials with minimum specified yield strength of 60,000 psi or 420 MPa and above, the selected consumables shall produce welds with tensile strength property not exceeding by more than 25% the specified minimum tensile strength properties, measured at the time of welding procedure qualification tests on an all-weld tensile test specimen taken from the weld metal.
- For carbon steel materials, inspection certificate type 2.2 test reports shall be submitted to the COMPANY for all consumables and shall be available at site at the time of welding. For all other materials, type 3.1.B inspection certificates shall be submitted to the COMPANY for all consumables and shall be available at site at the time of welding.
- Filler metal for welding similar materials shall be as shown in Table 1.
- 9% Ni materials shall be welded using filler materials complying with classifications of AWS A5.1. Guaranteed strength levels are required for mechanical design.
- Welds joining carbon steel materials of different grades shall give the same strength as that specified for the higher grade of material and shall have ductility and notch toughness properties equal to the higher values specified for the grades of steel being joined.
- In welding processes other than SMAW or FCAW, the bare filler wire shall contain all the alloying elements; no elements shall be added via the flux.
- All welding products shall be used within the limits recommended by their Manufacturer and the welding variables used for fabrication shall be within the range used for the procedure qualification.

5.1.2. Supply

- Low hydrogen electrodes and fluxes shall be supplied in sealed moisture proof containers.

5.1.3. Storage

- Electrodes, filler wires and fluxes shall be kept clean, dry and properly stored according to Supplier's recommendations. Storage of open containers shall be in accordance with Supplier's instructions but at a minimum temperature of 75°C in any case. No electrodes, filler wires or fluxes that are damaged, damp, greasy or oxidised may be used.

- CONTRACTOR's facilities shall include a temperature and humidity controlled welding consumable and flux store, including holding and drying ovens.
- Low hydrogen electrodes and fluxes shall be placed in a holding oven and held at a minimum temperature of 150°C for at least one hour prior to use. Welding electrodes shall be issued for
- production from holding ovens only and shall be placed in heated quivers capable of maintaining a minimum temperature of 70°C. After four hours, the remaining unused electrodes in the quivers shall be placed in a drying oven and held at 250-300°C for four hours or as recommended by the Supplier if at a higher temperature, and then transferred to a holding oven prior to reissue.
- Low hydrogen electrodes which have been in direct contact with water shall be definitely rejected and removed from the fabrication site.

5.1.4. Recycling of the fluxes

- Recycling of the flux is allowed under the following conditions:
 - a. Recycled flux shall be mixed with new flux in ratio of 1:1 (one volume of recycled flux with one volume of new flux)
 - b. A procedure shall be submitted to the COMPANY for acceptance detailing all the operations implemented: recovery of the used flux, sifting, mixing, drying, storage, etc.).

5.2. Backing Rings and Consumables Inserts

- Permanent backing rings and consumable inserts shall not be used. Temporary backing rings may be used after approval of the COMPANY and only if they are made of the same material that has to be welded; in such a case, at least the two first layers of passes shall be gouged out after completion of the weld, and rewelded. Other technique shall require the specific COMPANY prior approval.

5.3. Filler Metal and Fluxes

- Generally, the Consumables shall conform to those specified in ASME section II part C or the equivalent in AWS listed below:
 - a. AWS A5.1 - Specification for carbon steel electrodes for shielded metal arc welding
 - b. AWS A5.5 - Specification for low alloy steel covered arc welding electrodes
 - c. AWS A5.17 - Specification for carbon steel electrodes and fluxes for submerged arc welding
 - d. AWS A5.18 - Specification for carbon steel filler metals for gas shielded arc welding
 - e. AWS A5.20 - Specification for carbon steel electrodes for flux cored arc welding
 - f. AWS A5.28 - Specification for low alloy steel filler metals for gas shielded arc

welding

g. AWS A5.29 - Specification for low alloy steel electrodes for flux cored arc welding

- All welding consumables shall be selected to produce welds with yield strength exceeding that specified for the parent material. Where steels with different specified properties are joined the weld metal yield strength shall match or exceed that of the higher strength grade.
- For sour service applications the deposited weld metal shall comply with the requirements of NACE MR0175.
- If low-hydrogen electrodes are selected the diffusible hydrogen content shall not exceed 10 ml/100 g in the resulting deposited weld metal.
- Where no ASME material specification exists for wire or rods, wire or rods of the same nominal composition as the base material with substantially neutral flux or inert gas may be used if they have been qualified in accordance with the applicable code and specification requirements.

6. Joint Preparation, Spacing and Alignment

6.1. Edge Preparation

- Weld bevels shall be suitable for the welding process to be used and shall be as defined in the applicable qualified welding procedure. The bevel root gap and offset of butted edges shall be as required by the applicable construction code. Whenever necessary, spacing tools shall be used to ensure proper root opening. Double sided weld joints shall be used whenever possible; joints designed in single-welded butt joints shall be such that full penetration is attained. Double sided weld joints must have the root pass entirely cleaned to sound metal by arc-air gouging, followed by grinding, prior to back-welding.
- Removable starting and stopping tabs shall be used for longitudinal welding where automatic welding processes are used.
- Weld bevels shall be made by machining, grinding or thermal cutting, and the surfaces shall be reasonably smooth and true.
- On manual flame cutting, metal shall be ground-off to sound, smooth surface to allow for proper welding.
- Materials which require preheat for welding shall be preheated in the same manner for thermal cutting or arc-air gouging.
- Arc-air gouging is only acceptable on carbon steel material. When arc-air gouging is used, the surface shall have all carburized and hardened surfaces removed by grinding, and all such areas shall be inspected by the CONTRACTOR's quality control department for conformity prior to welding.
- All surfaces and edges to be welded shall be smooth, uniform and free from cracks, tears, gouges and other discontinuities which could adversely affect the quality or strength of the weld.

6.2. Cleaning

- Surfaces to be welded (weld levels and adjacent surfaces) shall be clean and free from paint, oil, dirt, scale, oxides and other materials detrimental to welding. Cleaning shall be done in a manner that will not lead to contamination of the weld or base metal. Only stainless-steel brushes and tools shall be used on stainless steels, nickel and non-ferrous materials. Grinding discs containing sulphur (iron sulphite) or other harmful components shall not be used on stainless steels, nickel or non-ferrous materials.
- Upon completion of each welding pass, the weld shall be cleaned of spatter, slag and flux deposits.
- Immediately before setting-up for welding and the bevel preparation of the edges, the surface of the edges shall be stripped of oxide films starting from at least 20mm from the edge by means of filing or stainless-steel wire brushing both internally and externally.

The edges shall further be cleaned to remove any trace of dirt, grease, etc., using a clean cloth or paper towel and a non-toxic solvent.

6.3. Tack Welds

- All tack welds shall be made in accordance with a previously approved welding procedure and shall be performed by qualified welders. Tack welds shall be of sufficient cross-section and length in order to avoid cracks, especially on high strength steel materials.

6.4. Temporary Welds

- Temporary welds for temporary handling attachments, lugs, etc. shall be minimized. When temporary welding is deemed necessary, welding shall be performed with the same approved procedures as for the main fabrication welding.
- The temporary welds shall subsequently be removed according to an approved procedure and ground flush with the base material, and then inspected by non-destructive methods (Magnetic Particle or Dye Penetrant). The method of removing attachments shall not injure the metal surface (i.e. they shall not be removed by hammering off).
- Any defect found during this inspection shall be removed and repaired by a qualified welder using welding procedures approved by the COMPANY.

7. Production Welding

7.1. General requirements

- Production welding may only start when qualified welding procedures and welders have been approved by COMPANY.
- Preparation for welding, types and sizes of electrodes, current amperages and voltages used, preheating requirements etc. shall be the same for production welding as those defined in the applicable qualified welding procedures.
- Adequate precautions shall be taken and suitable equipment shall be available on site in order to protect the welds (and welders) from adverse weather conditions (rain, wind, etc.) at the time of welding. If weather conditions become too severe, welding operations shall be stopped. No welding shall be done on wet base materials. Moisture shall be removed from surfaces to be welded by preheating to 50°C prior to welding. Welding shall not be carried out under heavy wind, unless appropriate shelter is provided for each welder/operator.
- All surfaces to be welded shall be visually inspected: they shall contain no laminations or other injurious defects.
- Filler rods for inert gas tungsten arc welding shall be cleaned immediately before use. Electrode wire for inert gas metal arc welding shall be protected from contamination during use and in particular between non-working periods.
- Peening of welds is not permitted.
- Structural attachment welds shall be continuous (skip welding not permitted). Fillet welds shall have a minimum leg length of 4 mm.
- Cleanliness shall be maintained after completion of welding. All stubs, rods, flux and foreign materials shall be removed from the equipment or piping.
- Stainless steel surfaces shall be restored to their corrosive resistant state. Chemical cleaning procedures, if applicable, shall be subject to the COMPANY approval.

7.2. Welding Processes

7.2.1. Approved welding processes

- All welds shall be made by either the Shielded Metal Arc (SMAW), Gas Tungsten Arc (GTAW), or Submerged Arc (SAW) processes, except that for Aluminium and Aluminium alloys, only the argon GTAW or GMAW processes shall be used. Any other welding process such as GMAW or FCAW (Flux Cored Arc Welding process) is not allowed except when authorised in relevant General Specifications.

7.2.2. Limitations and requirements

7.2.2.1.SMAW

- Use of SMAW in the downhill direction is restricted to utility lines; any other vertical welding shall be done vertical up.
- The maximum width of weave shall not exceed two times the bare electrode (core) diameter.

7.2.2.2.SAW

- The maximum individual layer thickness for submerged arc welds shall not exceed 10 mm. Recycling of fluxes shall not be permitted for use on materials other than carbon steels.

7.2.2.3.GMAW

- Gas metal arc welding with solid wire electrodes may only be used in the spray transfer current range. Short circuit transfer process shall not be used.

7.2.2.4.GTAW

- Thoriated tungsten shall not be used for the non-consumable electrode when welding Aluminium or Aluminium alloys.

7.2.3. Specific requirements

- Single-side welded butt joints in carbon-molybdenum, chromium-molybdenum steels, stainless steels, nickel alloys and welds located in pump or compressor suction or lubrication oil systems, shall have the root pass and hot pass stainless made by the GTAW process with an appropriate back gas shielding protection. When the GTAW or GMAW processes are used for welding carbon, carbon-molybdenum and low-alloy chromium molybdenum steels with a chromium content not exceeding 1.5%, in other applications than above, the requirement for back gas shielding may be waived, subject to the COMPANY prior approval.

7.3. Welding technique

7.3.1. Welding sequences

- The CONTRACTOR shall provide and implement all necessary supervision to ensure that the planned sequences are observed.

- The sequences in assembling, joining and welding the various parts to be welded shall be scheduled in order to minimise distortion, warpage, shrinkage and accumulations of residual joint stresses in each part of the piping systems or of the equipment being welded.
- Insofar as practicable, all welds shall be deposited in a sequence that will balance the applied heat of welding while welding is in progress.
- The direction of the general progression in welding parts of piping systems shall be from points where the parts are relatively fixed in position with respect to each other towards points where they have a greater relative freedom of movement.
- Joints expected to have significant shrinkage should usually be welded before joints expected to have lesser shrinkage. They should also be welded with as little restraint as possible.
- In making welds under conditions of severe external shrinkage restraint, the welding shall be carried-out continuously to completion or to a point that will insure freedom from cracking before the joint is allowed to cool below the minimum specified preheat and interpass temperatures.
- Where residual joint stresses are not desirable and cannot be avoided, suitable postweld heat treatments shall be performed. The heat treatments shall be made taking into account the recommendations of this Specification.
- Welded joints shall be made by completing each layer before the following layers are deposited.

7.3.2. Heat input control

- The heat input, in conjunction with the maximum preheat and interpass temperatures, shall be restricted to the maximum values shown in the relevant WPS and PQR for the following materials:
 - a. Austenitic and austenitic-ferritic stainless steels
 - b. Nickel alloys
 - c. Quenched and tempered carbon steels
 - d. Carbon steels when impact tested at temperatures of minus 30°C and below.
- The following parameters shall be carefully checked:
 - e. Preheat temperature
 - f. Minimum and maximum interpass temperatures
 - g. Amperage and voltage
 - h. Welding travel speed.
- The CONTRACTOR shall have available at the work site suitable equipment to measure these important variables (e.g. contact thermometer, etc.).
- Accuracy of parameters readings:
 - a. Temperature $\pm 10^{\circ}\text{C}$

- b. Amperage and voltage $\pm 5\%$.

7.4. Weld Contour and Finish

7.4.1. Flux and slag removal

- Weld beads shall be contoured to permit complete fusion at the sides of the bevel and to minimise slag inclusions. Flux, slag, and weld spatters shall be completely removed from weld beads and from the surface of completed welds and adjoining base material. The flux removal shall be done in a manner that will not contaminate or overheat the weld or adjoining base material.

7.4.2. Weld reinforcement

- Weld reinforcement and finish shall be as required by the applicable Code, except that undercutting of the base metal shall not be permitted. Internal protrusion shall not exceed the reinforcement thickness specified by the applicable Code. Weld finish shall not impair the interpretation of NDT results.

7.4.3. Weld surfaces

- Weld surfaces shall be free of cracks, porosity, slag inclusions and other defects indicative of poor workmanship.

7.4.4. Arc strikes

- Arc strikes outside weld bevels shall be avoided. Should this occur, however, the deposit shall be carefully removed by grinding and the area shall be 100% examined by magnetic particle (on magnetic materials) or dye penetrant (on non magnetic materials). Repairs shall be in accordance with this Specification.

7.5. Additional Requirements for Corrosion Resistant Overlays

- For weld overlays, there shall be no less than two layers of corrosion-resistant weld metal. However, a single layer overlay may be submitted to the COMPANY approval together with the use of the ESW (Electro Slag Welding) process exclusively.
- For stabilised or low-carbon 18/8 type Cr-Ni austenitic stainless steel cladding, the first layer of deposited weld metal shall be 309 type Cr-Ni austenitic stainless steel. Subsequent layers of deposit shall be of the matching grade of the clad material.
- When joining integrally clad plate;
 - a. The clad layer shall be stripped for a minimum distance of 10 mm from the bevel and the base material shall be etched with nitric acid to assure complete removal of cladding material
 - b. Removal of cladding shall not reduce the base material thickness below the design thickness
 - c. Preparations of local repair cavities in overlay welds which penetrate the base material more than 10% in thickness or 3 mm, whichever is the smaller, shall have the base metal rewelded with a relevant approved welding procedure.

8. Preheating

The following requirements shall apply in addition to those specified in the applicable Code. In case of discrepancy, the more stringent requirement shall apply, unless otherwise agreed by the COMPANY prior to qualification of the welding procedures.

- Preheat temperatures higher than those specified herein may be required during the qualification tests in order to meet contractual requirements such as hardness levels, or to prevent from cold cracking (particularly in the case of highly restrained welds).
- The preheating which has been qualified in the applicable WPS shall apply not only to welding, but also to all thermal operations such as thermal cutting, arc-air gouging, and also tack- welding.
- In weld assembly of dissimilar or different base materials, preheat temperature shall be the highest of those which are requested for each material.
- For carbon steel and low alloy carbon steels, preheat temperatures shall not be less than:
 - a. 50°C: When only moisture removal is required. When moisture removal is not applicable, minimum temperature of metal shall be 0°C for low hydrogen electrodes and 10°C for other types
 - b. 100°C: When the base material thickness exceeds 25 mm, or when the carbon equivalent exceeds 0.420.
- Note: Carbon equivalent CE is to be calculated from the IIW formula.
- For Aluminium alloys, preheat shall be as follows:
 - a. Prior to welding, the joint shall be heated uniformly to a temperature of 90°C but not more than 120°C
 - b. The above preheat temperature shall be maintained during all welding operations. On heavy sections, it is important that preheat be allowed to "soak" evenly throughout the base metal, so that the through metal temperature is 95-120°C. It is equally important that these temperatures (for preheat and interpass) are not exceeded
 - c. Should the welding be interrupted, the joint preheat temperature shall be maintained if possible. In any case, cooling under insulation is required, and re-heating at original temperature is necessary before welding resumes.
- For austenitic stainless materials, preheating is not required.
- The required minimum preheat temperature shall be applied to the whole length of joint to be welded and shall be maintained until the weld is completed. Application of preheat when using gas burners shall avoid deposition of carbon. The minimum preheat temperature shall be established for a sufficient distance either side of the joint to avoid unsatisfactory thermal gradient. This distance shall not be less 75 mm each side of the joint.
- The preheat temperature shall be measured on the face opposite to that being heated when possible. When this is not possible, allowance shall be made for temperature equalisation,

- e.g. remove heat source and allow period of one minute for each 25 mm thickness of material to elapse, before measuring temperature.
- Measurement of preheat and interpass temperatures shall be done by temperature indicating crayons, contact pyrometer or other approved method of temperature measurement. Whatever method to be used shall be stated on the welding procedure qualification and subject to approval by the Engineering Company. Maximum and minimum temperatures shall be specified. Temperature indicating crayons shall not be applied to surfaces where weld metal is to be deposited.

9. Postweld Heat Treatment

- Postweld heat treatment shall be in accordance with the applicable Code. Additionally, the following shall apply;
 - a. The necessity for post weld heat treatment shall be determined from the applicable equipment design code or standard. In the absence of specific requirements from such codes or standards, the requirements of ASME B31.3 shall be followed.
 - b. Stress relieving of welds in pipes of API Spec 5L Grades X60 and above, shall be carried out in the range 560 - 600 °C. Stress relieving of welds in all other grades of carbon steel shall be carried out in the range 580 - 620 °C. The holding time shall be 2.5 minutes per millimetre wall thickness, with a minimum time of one hour. The heating and cooling rate shall be a maximum of 300 °C per hour.

10. Radiographic Inspection

- Radiographic inspection of welds may be performed with gamma rays for piping only. In all other applications, X-rays shall be used exclusively, unless otherwise agreed in writing by the COMPANY.
- Films shall be carefully identified. The markings shall not be placed on welds and shall include weld number, line identification and welder's symbol.
- Radiographic inspection shall be done by X-rays as per ISO 5579, or ASTM E 1815 of better. The choice of other standard such as the European standard EN 584-1 category C3 by CONTRACTOR may be approved by COMPANY if Proved to be better than the afore mentioned. Subject to written approval by COMPANY, radiography may be performed using gamma rays.
- Sensitivity of films shall be verified by the use of wire-type Image Quality Indicator (IQI) placed transversely to the weld on the source side, for each exposure.
- Single-film viewing shall be used, and the film density shall be in the range of 2.5 to 3.5. For those instances where the variable thickness makes single film impracticable, with COMPANY approval, double film viewing may be used. For the double film technique, the film density shall be in the range of 2 to 3.5 for the double film combination or each individual film. Higher densities (up to 4.0) may be acceptable if adequate viewing and satisfactory interpretation of higher density film are permitted by the viewing equipment.
- The geometrical unsharpness shall not exceed 0.2 mm (Pressure vessels and piping shop fabrication) or 0.5 mm (Site assembly).
- Radiography of welds in pipe having a nominal diameter of 3" or less may be performed by the elliptical projection technique. At least two separate exposures are required at locations 90° apart
- Where there is no internal access, radiographs of welds in pipe shall be double wall technique with a minimum of two shots up to 4" in diameter and three shots over 4" in diameter and only that portion of the weld on the film side of the pipe (opposite to the radiation-source side) shall be interpreted. Because of the variation in pipe diameters, wall thicknesses and source-to-film distances, it may be necessary to take more than the minimum number of radiographs to properly examine the entire circumference of a weld.
- Magnetic particle examination shall be performed by the electro-magnet method. The prods method shall not be used because this equipment could produce arc strikes on the pieces to be tested.
- The extent of inspection shall be in accordance with the contractual requirements and shall not be less than the following:
 - a. All welds shall be 100% visually inspected after completion and before any the NDT technique is applied.

- b. Welds of a size or type preventing conclusive radiograph images (this does not apply to welds in pressure retaining service including nozzle welds), such as some types of branch connections and fillet welds, shall be given a magnetic particle examination. For non-magnetic materials and nickel alloys, a liquid penetrant examination shall be used. Internal surfaces shall be examined where accessible.
- When welds in pressure-containing equipment of chromium-molybdenum materials are subject to 100% radiography, the welds shall also be examined inside (when accessible) and outside by the magnetic particle method after post weld heat treatment.
- The attachment welds between structural components and pressure parts shall be either dye penetrant or magnetic particle examined. For heat-treated components, this shall be done after post weld heat treatment.
- All pressure-containing equipment, irrespective of material, shall have all nozzle and reinforcing pad attachments and attachments to pressure parts welds examined by the magnetic particle or liquid penetrant method. Inspection shall be performed before and after post weld heat treatment both outside and inside where accessible.
- For spot radiographed equipment, radiographic examination shall not be less than 10% of the weld length as well as 100% of well crossings.
- When radiographic examination is required and is not practicable, ultra-sonic examination plus magnetic particle inspection may be considered as an alternate subject to the COMPANY approval. Dye penetrant inspection may also be used to replace MPI when deemed necessary, subject to the COMPANY approval.
- For clad welds or weld overlays, all welded surfaces shall be 100% inspected by dye penetrant inspection. Linear defects or pin holes are not acceptable.
- Tungsten inclusions in welds (when using the GTAW process) are not permitted and shall be repaired even if small in size or in quantity, if located in the root pass.

11. Rejection and Repair

- Indications or imperfections that are outside the limits of the applicable codes and standards, this Specification, Project Specifications or other requirements stated on the Purchase Order, shall be cause for rejection and the CONTRACTOR shall take such remedial action as is necessary to secure acceptance. Such re-work shall be subject to the COMPANY approval.
- Complete repair of a weld shall include removal of the weld, bevelling of new edges, and re- welding.
- When welding is judged to be unsatisfactory, the welder or welder operator responsible for the work shall be suspended from welding and all his work examined by non-destructive means. Welding found to be unacceptable shall be repaired. The welder or welder operator may be re- assigned only after additional training, the completion of satisfactory re-qualification tests, and with the approval of the COMPANY.
- All repairs shall be documented and shall be included in the CONTRACTOR's data book. Documentation on repairs shall include the following (for each repair):
 - a. Location and layout of defect
 - b. Nature of defect and means of detection used (NDO)
 - c. Repair work actually performed (defect removal method, WPS used)
 - d. NDE performed and results obtained during and after repair.
- Repairs shall be advised to the COMPANY; the repair method, the welding procedure and welder qualifications, etc., shall be in accordance with this Specification and shall be approved by the COMPANY before any repair being initiated. This approval must be based on qualification testing (PQR) supporting a specific WPS for each type of repair.
- Welds containing cracks shall be subject to additional non-destructive testing (Ultrasonic or Magnetic Particle) and then, the extent of repair shall be decided by the COMPANY. Repair welding will only be permitted after consideration of the nature and cause of cracking.
- Unacceptable defects shall be removed by chipping, grinding, machining or air-arc gouging. Where air-arc gouging is used, all carbon, cooper and other debris, including carburized metal, shall be removed by grinding or other mechanical methods approved by the COMPANY. Oxygen gouging of quenched and tempered steels or other high strength steels is not permitted.
- For partial repairs, the cut-out portion shall be sufficiently deep and long to remove the defect. At the ends and sides of the cut, there shall be a gradual taper from the base of the cut to the surface of the weld metal. The width and profile of the cut shall provide adequate access for rewelding.
- Special care shall be taken to remove weld defects located at the root in order to obtain an acceptable root gap.

- Prior to starting the repair, the repair grooves shall be examined by dye penetrant method, to ensure that all defects are removed.
- Preheating (where required) and interpass temperatures shall be maintained during all weld repairs.
- Repairs on already post weld heat-treated components shall be subject to a new post weld heat treatment under the same conditions and rules.
- All repaired welds shall be re-inspected in accordance with this specification at the CONTRACTOR's expense. Moreover, additional radiographs of two welds previously performed by the same welder shall be required at the CONTRACTOR's expense. These additional two welds shall be selected by the COMPANY.
- For 3 ½ Ni materials, as well as for austeno-ferritic duplex stainless steels, a second repair of the same area on a given weld shall not be permitted: should a defect be repaired on a previously repaired weld area, the whole weld shall be cut and rewelded after having restored the base materials to their initial condition (i.e. remove heat affected zones, damaged surfaces, etc.).
- For all other welds, only two repairs shall be permitted on the same area of a given weld: should a third repair become necessary, then the whole weld shall be cut as per above.
- In any case, the WPS used for these repairs shall be qualified on the same number of succession repairs actually performed.

12. Welder Identification

- In field welding, each qualified welder or welding operator shall have an identification symbol assigned to him and, unless specified otherwise, shall permanently mark each pressure weld with his identification symbol. If more than one welder welds a joint, each shall apply his symbol in a manner to indicate the part of the joint he welded.
- For shop welds, the CONTRACTOR shall keep a record of the welder(s) employed on each joint.

13. Records of Inspection

- The CONTRACTOR shall keep a complete and accurate current record of all inspections and their results. These records shall be available for examination by the COMPANY at all reasonable times.